DAY 09 — LINUX FUNDAMENTALS & TERMINAL NAVIGATION

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THEORY — 20 Questions with Answers

1. What is Linux?

Linux is an open-source operating system family used in servers, cloud systems, networking, cybersecurity, development and embedded devices.

2. What is a Linux distribution?

A Linux distribution is a package that includes the Linux kernel, system utilities, package manager, applications and sometimes a desktop environment.

3. Name four Linux distributions.

1. Ubuntu
2. Debian
3. Fedora
4. Kali Linux

4. What is Kali Linux?

Kali Linux is a Linux distribution specially designed for security testing and cybersecurity-related work.

5. What is a terminal?

A terminal is an interface that allows users to communicate with the Linux operating system by typing commands.

6. What does the whoami command do?

The whoami command displays the username of the currently logged-in user.

7. What does hostname display?

The hostname command displays the name of the computer or system.

8. What does pwd mean?

PWD stands for Print Working Directory. It displays the current directory in which the user is working.

9. What is the purpose of ls?

The ls command is used to list files and directories in the current location.

10. Explain the difference between ls, ls -l and ls -a.

- ls – Displays normal files and directories.
- ls -l – Displays detailed information such as permissions, owner, file size and date.
- ls -a – Displays all files, including hidden files.

11. What does cd do?

CD stands for Change Directory. It is used to move from one directory to another.

12. What does cd .. do?

The cd .. command moves the user one level up to the parent directory.

13. What does cd ~ do?

The cd ~ command takes the user directly to their home directory.

14. What does cd / do?

The cd / command takes the user to the root directory of the Linux filesystem.

15. What is the Linux root directory?

The root directory (/) is the top-level directory of the Linux filesystem. All other files and directories are organized under it.

16. What is the difference between the root directory and the root user?

- Root Directory (/): It is the top-level directory of the Linux filesystem.
- Root User (root): It is the privileged administrative user with full system access.

17. What is an absolute path?

An absolute path is the complete path of a file or directory that always starts from the root directory (/).

Example: /home/student/Documents/report.txt

18. What is a relative path?

A relative path specifies the location of a file or directory based on the current working directory.

Example: Documents/report.txt

19. How are hidden Linux files commonly identified?

Hidden Linux files commonly begin with a dot (.) before their filename. They can be viewed using the ls -a command.

Example: .bashrc

20. Why are Linux skills important for cybersecurity?

Linux skills are important because cybersecurity professionals use Linux terminals to inspect systems, analyze network activity, review logs, run security tools, investigate incidents and manage servers.

PRACTICAL SCREENSHOTS:

- **whoami**
- **hostname**
- **pwd**
- **ls**
- **ls -l**
- **ls -a**
- **cd /**
- **cd ~**

